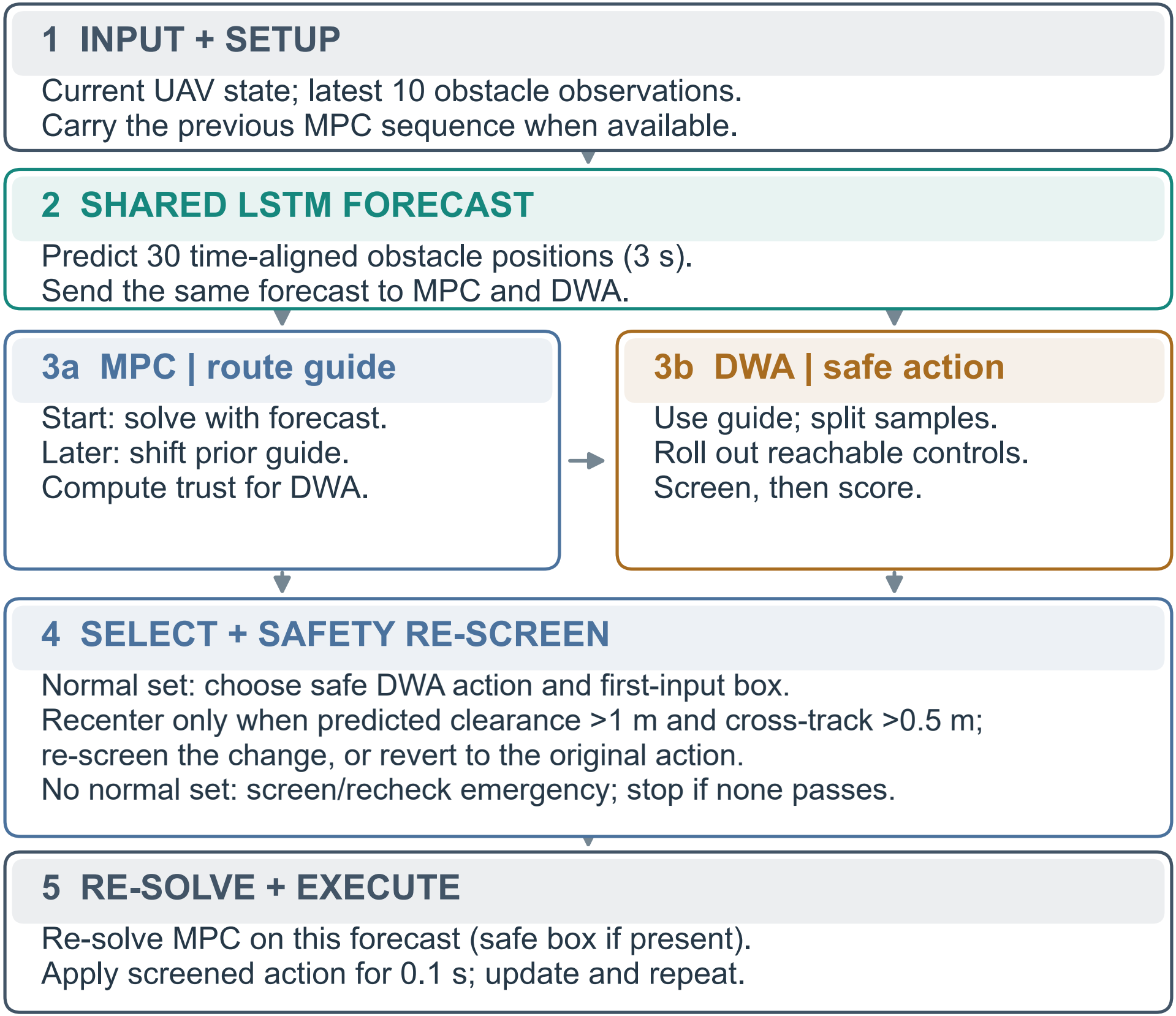
1 INPUT + SETUP
Current UAV state; latest 10 obstacle observations.
Carry the previous MPC sequence when available.
2 SHARED LSTM FORECAST
Predict 30 time-aligned obstacle positions (3 s).
Send the same forecast to MPC and DWA.
3a MPC | route guide
Start: solve with forecast.
Later: shift prior guide.
Compute trust for DWA.
3b DWA | safe action
Use guide; split samples.
Roll out reachable controls.
Screen, then score.
4 SELECT + SAFETY RE-SCREEN
Normal set: choose safe DWA action and first-input box.
Recenter only when predicted clearance >1 m and cross-track >0.5 m;
re-screen the change, or revert to the original action.
No normal set: screen/recheck emergency; stop if none passes.
5 RE-SOLVE + EXECUTE
Re-solve MPC on this forecast (safe box if present).
Apply screened action for 0.1 s; update and repeat.